Topological Quantum Matter

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Abstract

This is a Lecture note on Prof. Qingrui Wnag's lecture in Topological Quantum Matter.

Contents

1 Lecture 1: Introduction and Motivation

This section serve as an

1.1 Toric Code Model

This is a CMT model on lattice, which is initially defined on a **torus**, and it can be used to contain information thus called **code**. This is in fact a $\mathbb{Z}_2$ gauge theory.

The model is defined on a square lattice, with on each **link** a spin-1/2 degree of freedom. The Hilbert space is then the tensor product of all the spin-1/2 Hilbert space on each link:

$$\mathcal{H} = \bigotimes_{\text{link}} \mathbb{C}^2 \tag{1}$$

It is interesting for giving following properties:

- The ground states of the model represent $H_1(T^2, \mathbb{Z}_2) = \mathbb{Z}_2 \times \mathbb{Z}_2$.
- There are 4 degenerate ground states, which are topologically protected. This means that the ground states cannot be change into each other by any local operation, and the degeneracy is robust against any local perturbation.

1.1.1 Model Setup

To define the model, we first define the Hamiltonian which is a operator:

$$H : \mathcal{H} \to \mathcal{H} \tag{2}$$

This Hamiltonian is defined can be constructed by some local operators on latticeusing the pauli matrices:

$$\sigma^i : \mathbb{C}^2 \to \mathbb{C}^2 \tag{3}$$

Now we can define the Hamiltonian as:

Definition 1.1 **Hamiltonian of Toric Code Model**

$$H = -\sum_{\{v\}} A_v - \sum_{\{p\}} B_p \tag{4}$$

where A_v and B_p are defined as:

- $A_v = \prod_{i \in v} \sigma_i^x$, where v is the set of links that are connected to the vertex v.
- $B_p = \prod_{i \in p} \sigma_i^z$, where p is the set of links that are on the boundary of the plaquette p.

Here is a diagram of the model:

Bibliography